

# WHEN "I'M FINE" ISN'T FINE

## Response Strategy Selection Under Acoustic-Semantic Emotional Incongruence in Supportive Dialogue

Yubo Qiu, Fiona Zhang / Carnegie Mellon University

### Motivation: Hidden Emotions in Conversation

#### "I'm fine." – said with a trembling voice

- People often **mask their true feelings** when speaking: saying positive words while their voice betrays negative emotions, or vice versa.
- Text-only ESC systems fail when speech tone contradicts words

MASKING

OVERSTATEMENT

"Everything is fine!"  
Positive words  
Voice: Sad, low energy

"This is absolutely terrible!"  
Negative words  
Voice: Cheerful, laughing

**SOTA:** E-THER shows that **mismatch awareness** helps empathy – but which strategy is best?

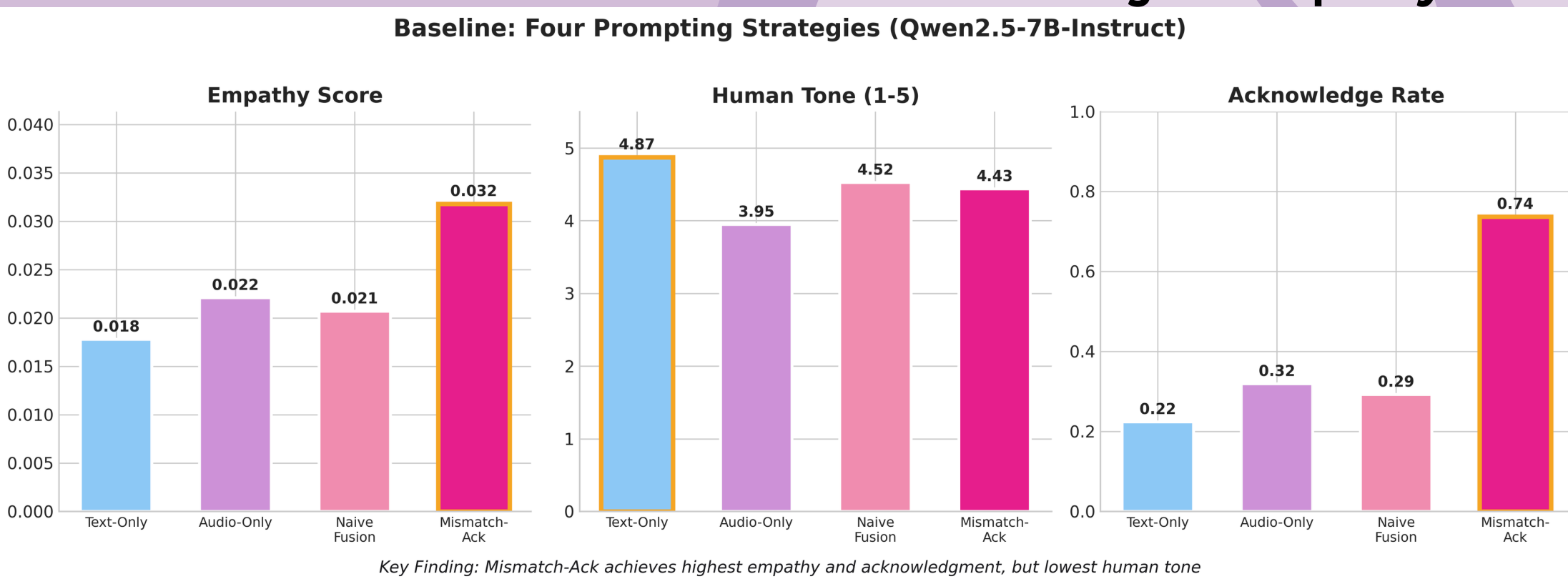
- Masking (hiding) → Providing Suggestions → decent
- Overstatement (joke) → Providing Suggestions → ?

### Baseline: Mismatch Awareness Trend

- Model: Qwen2.5-7B
- Dataset: MELD 232 samples
- Different model/data from E-THER, replicates trend

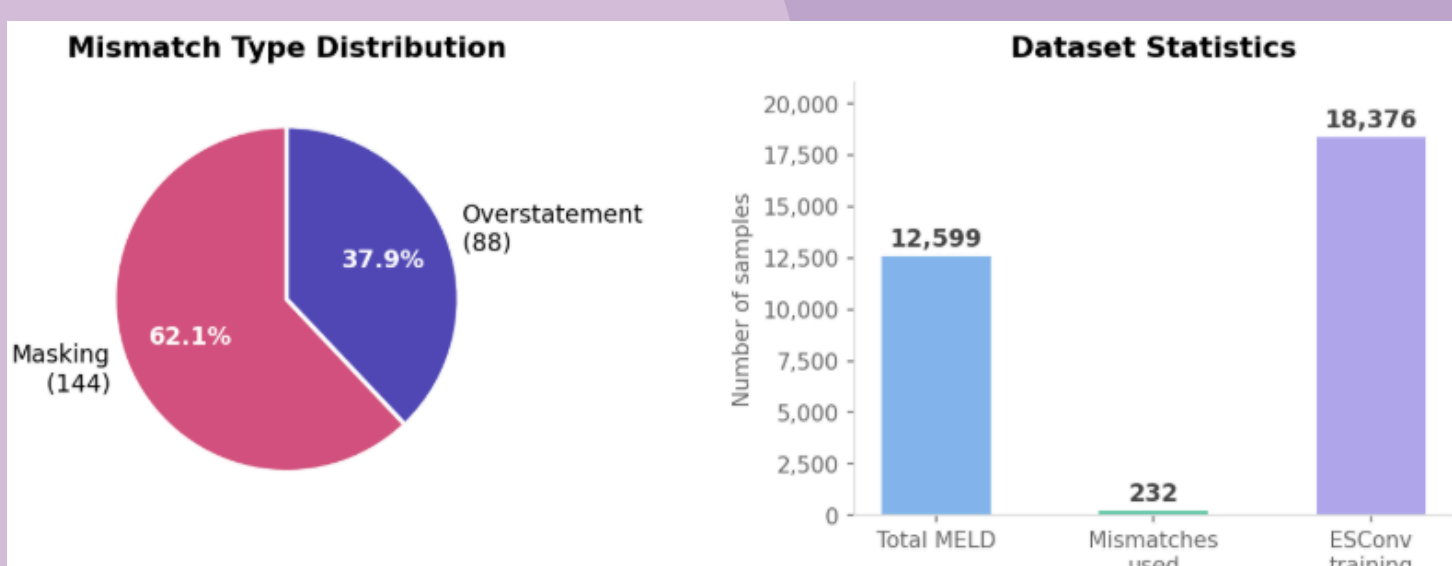
| Strategy     | Description                     | Example Response                                                                                        |
|--------------|---------------------------------|---------------------------------------------------------------------------------------------------------|
| Text-Only    | Respond to words alone          | "Glad to hear you are fine!"                                                                            |
| Audio-Only   | Respond to voice emotion        | "You sound like you're struggling."                                                                     |
| Naive Fusion | Mention both without insight    | "You said X and sound Y."                                                                               |
| Mismatch-Ack | Explicitly acknowledge conflict | "You say you're fine, but something in your voice suggests otherwise. Would you like to talk about it?" |

Four Conditions: **More mismatch awareness → higher empathy.**



### Setup:

- 232 mismatch** samples: **144 masking** + **88 overstatement** acoustic-semantic incongruent utterances from the MELD dataset
  - Multimodal EmotionLines Dataset from Friends TV series
- ESConv LoRA fine-tune** on **Qwen2.5-7B**
  - 18,376 turns, 3 epochs, H100
- 6 strategies** × each sample
  - within-experiment** no-strategy baseline
- 5 metrics:** RoBERTa empathy scorer + LLM judges for **human tone, acknowledgment, emotion correctness, and appropriateness**



#### Mismatch Detection Method

- Audio emotion: MELD ground-truth labels
- Text sentiment: DistilRoBERTa classifier (Hartmann, 2022)
- Conflict detection: When audio and text sentiment polarities disagree

### Additional Baseline Experiment: Audio Language Model Fine-Tuning

- Base Model: Qwen2-Audio-7B-Instruct
- Method: QLoRA (Parameter-Efficient Fine-Tuning)
- LoRA rank (r): 16 | LoRA alpha: 32
- Training Data: 982 samples
  - (441 real + 541 synthetic via ESConv few-shot)

**3.3x Improvement** in Acknowledgment Rate from ALM baseline

**MASKING - Baseline**

User: "I'm okay, really."  
Voice: Sadness

Response: "That's good to hear! Is there anything you'd like to talk about?"  
X Misses emotional cue

**MASKING - Fine-tuned**

User: "I'm okay, really."  
Voice: Sadness

Response: "I hear you saying you're okay, but something in your voice makes me wonder if there's more going on..."  
✓ Acknowledges mismatch

**OVERSTATEMENT - Baseline**

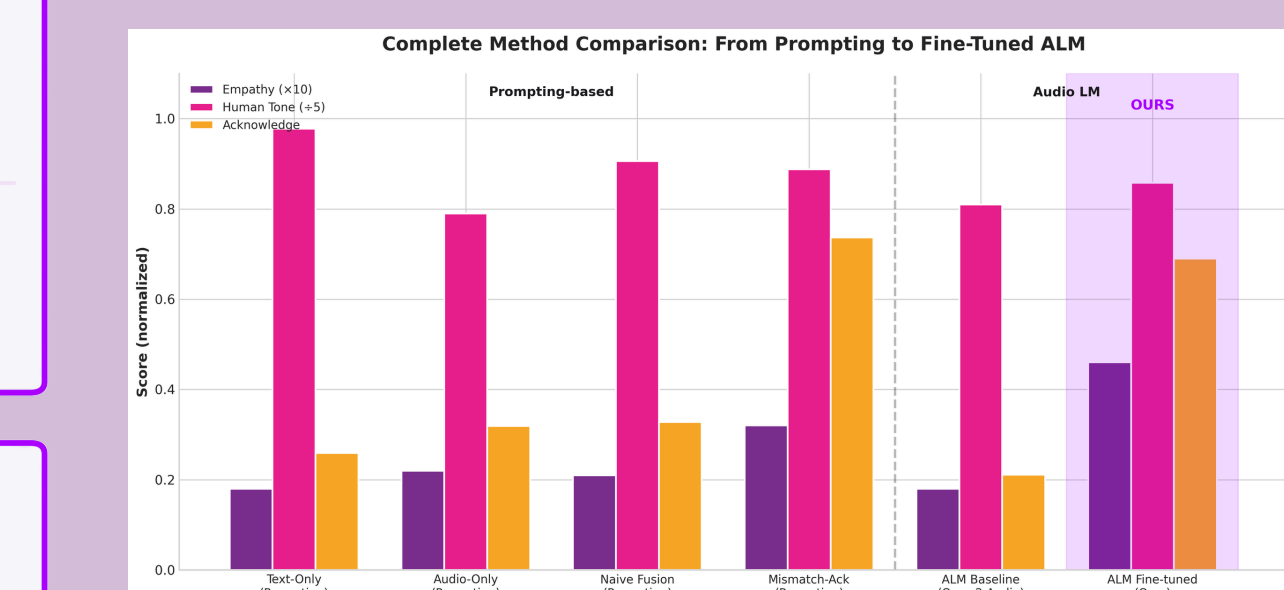
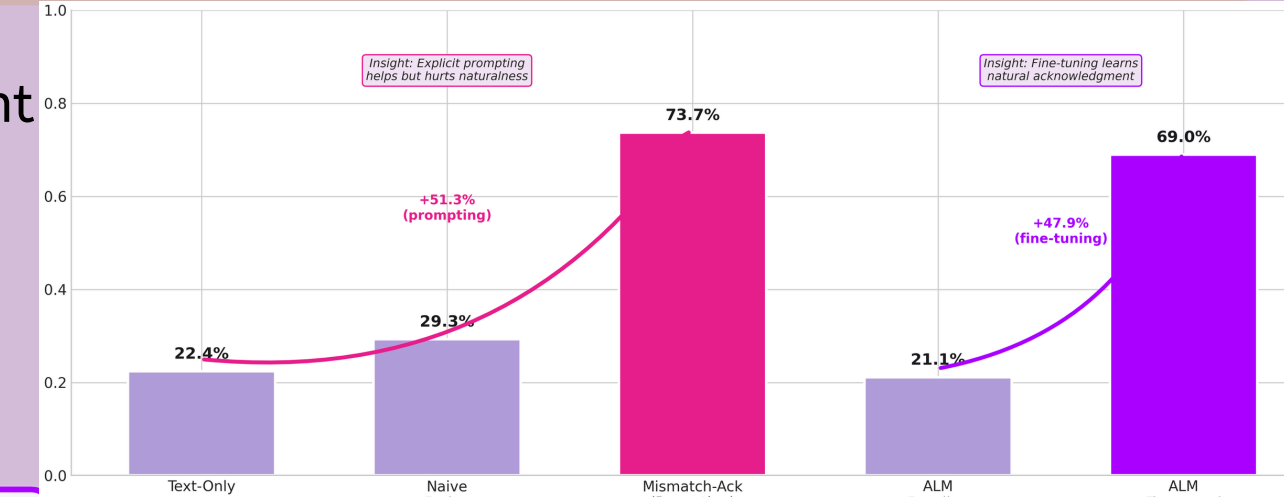
User: "This is the worst day ever!"  
Voice: Joy

Response: "I'm so sorry you're having such a hard day."  
X Takes words literally

**OVERSTATEMENT - Fine-tuned**

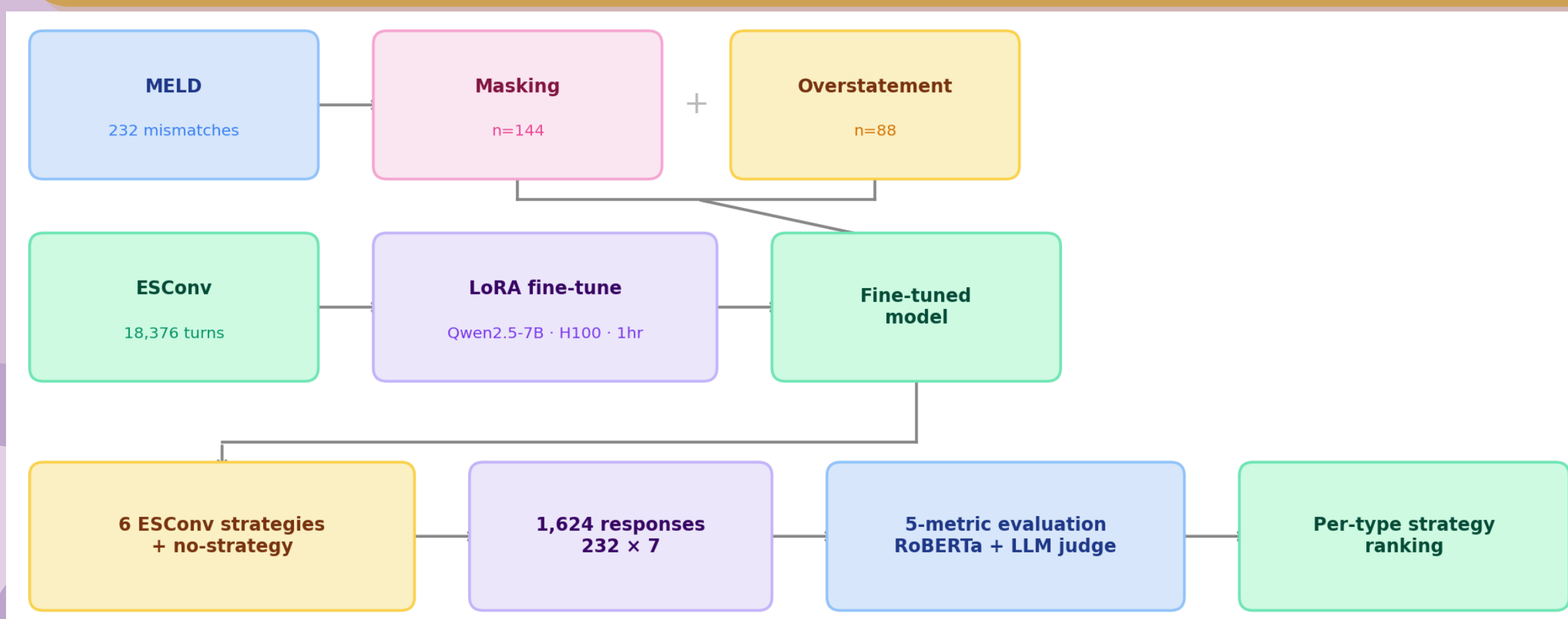
User: "This is the worst day ever!"  
Voice: Joy

Response: "It sounds like you're venting a bit! What's really going on?"  
✓ Recognizes exaggeration



Achieves **highest empathy** while maintaining acknowledgment comparable to explicit prompting, but with more natural responses.

### Pipeline



- Why within-experiment: **fine-tuning changes model behavior (reduces Ack)**, so external baseline not directly comparable
- All conditions controlled:** same model, same prompt structure, only strategy varies

### Results: Overall (n=232)

#### Masking

n=144

| Strategy                    | Empathy | Tone | Ack   | EC(0/1) | EC(1-5) | Ap(0/1) | Ap(1-5) |
|-----------------------------|---------|------|-------|---------|---------|---------|---------|
| Question                    | 0.0200  | 2.99 | 0.062 | 0.007   | 1.98    | 0.167   | 2.28    |
| Reflection of Feelings      | 0.0198  | 2.56 | 0.062 | 0.111   | 2.11    | 0.285   | 2.28    |
| Affirmation & Reassurance   | 0.0214  | 2.95 | 0.021 | 0.042   | 1.92    | 0.243   | 2.26    |
| Restatement or Paraphrasing | 0.0202  | 2.75 | 0.028 | 0.056   | 2.03    | 0.257   | 2.33    |
| Providing Suggestions       | 0.0231  | 3.11 | 0.056 | 0.028   | 2.06    | 0.285   | 2.42    |
| Self-Disclosure             | 0.0147  | 2.64 | 0.028 | 0.014   | 2.08    | 0.264   | 2.32    |
| No Strategy                 | 0.0166  | 2.91 | 0.042 | 0.021   | 1.83    | 0.208   | 2.15    |
| Random Strategy             | 0.0196  | 2.82 | 0.014 | 0.042   | 2.01    | 0.299   | 2.28    |

Legend:  = best among 6 strategies per metric  
dashed line separates within-experiment baseline (no strategy / random)

#### Overstatement

n=88

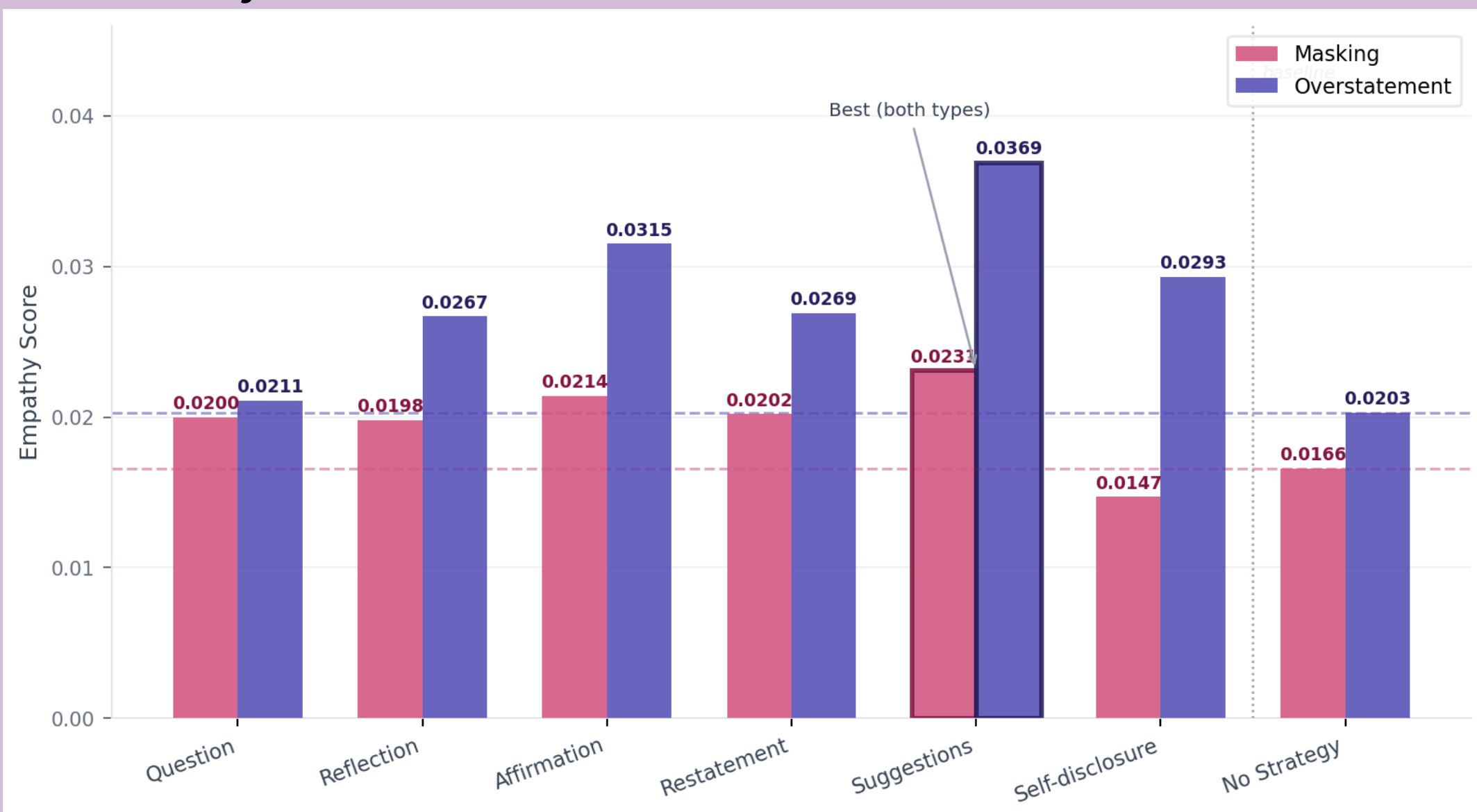
| Strategy                    | Empathy | Tone | Ack   | EC(0/1) | EC(1-5) | Ap(0/1) | Ap(1-5) |
|-----------------------------|---------|------|-------|---------|---------|---------|---------|
| Question                    | 0.0211  | 3.10 | 0.068 | 0.011   | 2.26    | 0.648   | 2.81    |
| Reflection of Feelings      | 0.0267  | 2.99 | 0.080 | 0.080   | 2.47    | 0.545   | 2.70    |
| Affirmation & Reassurance   | 0.0315  | 3.25 | 0.034 | 0.045   | 2.41    | 0.534   | 2.76    |
| Restatement or Paraphrasing | 0.0269  | 3.24 | 0.045 | 0.057   | 2.38    | 0.659   | 2.81    |
| Providing Suggestions       | 0.0369  | 3.06 | 0.068 | 0.011   | 2.39    | 0.500   | 2.83    |
| Self-Disclosure             | 0.0293  | 3.01 | 0.023 | 0.023   | 2.41    | 0.511   | 2.82    |
| No Strategy                 | 0.0203  | 3.10 | 0.045 | 0.034   | 2.27    | 0.455   | 2.82    |
| Random Strategy             | 0.0252  | 3.05 | 0.057 | 0.034   | 2.35    | 0.557   | 2.84    |

Legend:  = best among 6 strategies per metric  
dashed line separates within-experiment baseline (no strategy / random)

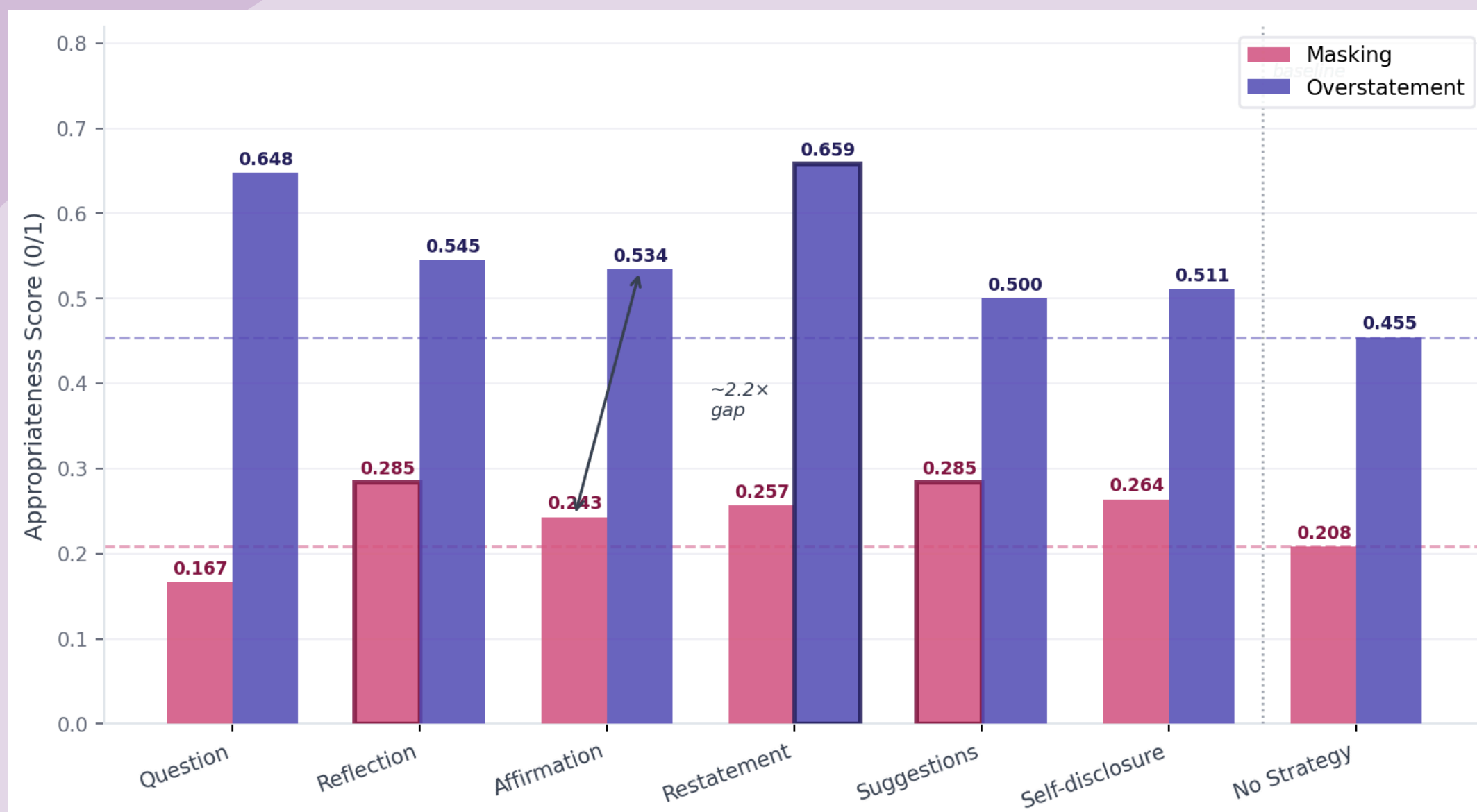
- EC and Ap are each evaluated at **two scales (0/1 binary and 1-5)** to examine whether finer-grained scoring reveals additional signal beyond binary judgment.

### Qualitative Analysis

- Strategy > No Strategy:** All 6 strategies outperform no-strategy on empathy overall. Random strategy (0.0231) only marginally better than no-strategy (0.0180) – targeted selection matters.
- Providing Suggestions leads by empathy for both types, but **Overstatement shows broader multi-strategy robustness** (Restatement highest Ap 0.659, Affirmation highest Tone 3.25).
- Overstatement consistently scores **higher than Masking** across all metrics. Masking is harder for the model.
- Ack score universally low** (0.03–0.08): ESConv fine-tuning reduces mismatch acknowledgment ability – strategy learning trades off with ack ability.



- Overstatement Ap scores 2–3× higher** than Masking across all strategies (0.50–0.66 vs. 0.17–0.28)
- Restatement best for Overstatement** (Ap 0.659) but only 0.257 on Masking – same strategy, opposite fit
- Providing Suggestions most robust across both types on empathy**
- Confirms: one strategy cannot serve both mismatch subtypes**



- Reflection of Feelings shows highest EC for Masking (0.111)** – only strategy with meaningful recognition of suppressed distress
- EC universally low across both types**, suggesting fine-tuned model struggles to identify underlying emotional state regardless of strategy
- EC and Ap rankings diverge** – high appropriateness does not guarantee correct emotion identification

### Conclusion

- Strategy-guided fine-tuning **outperforms no-strategy baseline**
- Type-specific patterns exist – both types **favor Providing Suggestions**, but **Overstatement benefits from multiple strategies**
- Overstatement more tractable** than Masking
- Fine-tuning trade-off: ESConv improves strategy adherence but **reduces mismatch acknowledgment**

### Future Work

- Train on mismatch-aware data to recover Ack ability
- Larger dataset with explicit mismatch × strategy annotations
- Explicit strategy-type pairing during fine-tuning